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Why We Keep Talking About NFTs

Pau Waelder

14 December 2021

[Aquí heu de situar el fitxer d'imatge cecile-alexia-harts-bust.jpg]

Alex John Ettl working on Cecile Alexia Hart's bust. 1920-1925 | Library of Congress |
No known copyright restrictions

In a very short period of time, NFTs have shaken up the art market and generated a financial bubble. The use of blockchain technology overcomes an old problem for net art, but also brings with it new controversies such as its environmental impact. We analyse the causes of this phenomenon and how it might develop in the future.

Collins Dictionary recently announced that it has chosen the acronym NFT as its word of the year 2021. True to its role, Collins provides a concise definition of this term that has had such an impact on the contemporary art market in recent months: non-fungible token: a unique digital certificate, registered in a blockchain, that is used to record ownership of an asset such as an artwork or a collectible.

Among the many ways in which NFTs have been described, the definition given by Collins has the value of clarifying a fact that has often led to confusion: an **NFT is a certificate**, not a type of artwork, nor a creative technique, nor an art movement. Just because something is minted on a blockchain as a non-fungible token does not make it a work of art, or even a work of digital art. NFTs do not really represent a transformation in the way artworks are created, nor do they embody a particular conceptual or aesthetic premise. It is not a technology that, in itself, provides artists with new means of expression or defines a visual style. Non-fungible tokens make it possible to certify the authenticity and ownership of a work of art in digital format (among other things), which is by nature infinitely reproducible, and this is their great contribution to the way in which some contemporary art is distributed, marketed and collected today.

Having clarified what an NFT is, one might wonder how a simple digital certificate could generate so much media buzz, disrupt the art market and fuel a financial bubble in barely eight months. It is a complex, multi-layered and nuanced story that does not lend itself to a simple explanation, let alone a complete one.

[Aquí heu de situar el fitxer d'imatge possession.jpg]

Screenshot of *Possession*, online collection of Auriea Harvey and Michael Samyn that hosts the work *If You Want To Clean Your Screen* (1998) by Olia Lialina.

NFTs resolve a problem present since the inception of net art

When, in the mid-90s, a small group of artists began to experiment with the resources offered by the World Wide Web and HTML code, they saw the possibility of freely disseminating their works without having to depend on curators or gallery owners. In 1998, the artist Olia Lialina had the idea of turning her website into an art gallery and sold one of her works to the artists Michael Samyn and Auriea Harvey, who received the files and posted them on their own website, in a section entitled *Possession*. The work was still publicly accessible, but now had other owners. However, not all digital art can be sold in this way, which has led many artists, including Mark Napier [1], Carlo Zanni [2] and Rafael Rozendaal [3], to consider how to market their work on the internet. The main problem has always been how to ensure the collector's ownership of a piece whose very nature is to be distributed and accessible to everyone.

The creation of Bitcoin in 2009 brought with it blockchain technology as a distributed and unalterable record – a resource that is particularly interesting for the art market, where the creation of value springs basically from the authenticity and scarcity of works of art. Starting in 2014, the first platforms were created that allow artists and creators to register intellectual property on a blockchain.

Speculative mania is fuelled by games, crypto investors and auction houses

For years now, many artists have been selling their digital works with certificates of authenticity on the blockchain, without this sparking any revolution. However, when, in 2017, non-fungible tokens were created using the ERC-721 standard on the Ethereum blockchain, the first projects that took advantage of its functionalities caused a sensation precisely because of the possibility of owning, and exchanging, something unique. CryptoPunks, a series of 10,000 character portraits generated by an algorithm, and CryptoKitties, a game that involves adopting and exchanging virtual cats, use combinatorics to obtain a large number of characters with personalised traits, which are made unique by being registered as non-fungible tokens. The growing interest in these and other projects, particularly among cryptocurrency investors, is being harnessed by the auction houses Christie's and Sotheby's, which have jumped on the bandwagon with sales of several artists, taking advantage of their privileged position in the art market to make substantial profits.

The spectacular prices achieved with the work of creators unknown to the art market, and in particular with pieces that anyone with 3D modelling skills can imitate, have triggered a rush to create NFTs among artists, illustrators, designers, programmers and opportunists alike. At the same time, a large number of cryptocurrency investors are looking to art as a defensive asset, without knowing (or perhaps even caring) exactly what art is or whether it is guaranteed to hold its value over time, and are buying art registered on the blockchain at prices well above what was seen previously on the market.

[Aquí heu de situar el fitxer d'imatge cryptopunk.jpg]

CryptoPunks, a series of 10,000 character portraits generated by an algorithm.

The NFT market reproduces the inequalities of the traditional market

The first wave of million-dollar sales led to a quick growth in activity in the NFT marketplaces created in 2017 and 2018, such as OpenSea, SuperRare and Nifty Gateway, based on the Ethereum blockchain. In 2020, two very different platforms were launched: Hic et Nunc (H=N), [4] a *rara avis* based on the Tezos blockchain, which established itself as the go-to for a broad community of artists, and Foundation, a marketplace specialising in digital art with a considerable portfolio of artists with an established track record in this medium.

Two art markets were soon established based on NFTs: one represented by platforms such as SuperRare and Foundation, which only accept artists by invitation and sell works in auctions at high prices using Ethereum, and the other represented by H=N and other platforms on Tezos such as Objkt, open to all types of artists who offer their works at more affordable prices. Another fundamental difference lies in the environmental impact of the Ethereum and Tezos blockchains. Minting non-fungible tokens on the Ethereum blockchain consumes enormous amounts of electricity, which in turn contributes to CO2 emissions and thus to global warming. The Tezos blockchain uses a different protocol which consumes much less energy, similar to that generated by sending a tweet for each NFT. While the actual energy cost is still a matter of debate, Ethereum is still more polluting than Tezos, but also potentially more profitable, as most investors operate in ETH.

Collecting (quality) art in NFTs will be commonplace

The NFT art market has developed at breakneck speed over the last few months, opening up a wide range of possibilities, within which some dominant trends and influential actors are emerging. One of these trends is the replacement of marketplaces that are open to all, in which each work is just another image in an infinite grid, by a **curated presentation of artworks with a specific discourse** conveyed in the format of an individual artist's project or a collective exhibition.

Another trend is the **identification of NFTs with a specific type of image**, specifically a 3D model or animation with a futuristic aesthetic, along the lines of Beeple's work, or abstract geometric compositions, also with three-dimensional effects, in the style of Pak. Finally, the current trend is towards the definitive assimilation of NFTs into the mainstream contemporary art market, as major galleries naturally incorporate this format into their offerings to collectors. Pace, one of the strongest galleries globally, has recently launched Pace Verso, a platform for selling NFTs that, notably, sets prices in dollars and accepts payment by credit card (as well as the main cryptocurrencies), in what can be interpreted as an invitation to "traditional" collectors who, while they don't share the enthusiasm of crypto investors, don't want to be left behind in the acquisition of digital art either.

An innovation cannot be said to have been assimilated by the art market until it is adopted by the galleries themselves. Now that we are at the point where both Pace and a growing number of other galleries have decided to sell NFTs, it can be considered that non-fungible tokens are going to be as common a format as silkscreen prints, digital prints, photographs or videos. It is also possible that the enthusiasm for these digital works is nothing more than a speculative bubble that hasn't stopped growing yet (remember that the dot-com bubble lasted five years) and that in a few years many collectors will find that their works are not worth what they paid for them.

But in any case, the resilience of the market and the involvement of its most powerful actors bodes a stable future for digital-format art and for records of authenticity and chain-of-ownership in blockchain, or other similar technologies. Whether under their oft-repeated acronym or under a different name, NFTs will continue giving us much to talk about.

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- [1] In 2002, Napier created The Waiting Room, a virtual online space shared exclusively by the collectors who acquire one of the 50 participations in the piece.
 - [2] For years, Zanni has been exploring the possibilities of marketing works of net art. One of his first pieces is Altarboy (2003-2004), which consists of a sculpture containing a server in which the work is hosted. To view it, the collector must connect the object to the internet, making the work publicly visible on a browser.
 - [3] Rozendaal creates pieces designed to be seen on a web browser and assigns them a domain name which makes each piece a unique object. A contract drawn up by the artist obliges the collector to keep the piece on the website, while adding their name to the piece's source code.
 - [4] The platform was closed down by its creator on 11 November 2021 and is now running on other servers. A brief history of Hic et Nunc can be found at Github.

[Aquí heu de situar el fitxer d'imatge paul-waelder.jpg]

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Waelder, Paul (14 december 2021). Why We Keep Talking About NFTs

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About NFTs

What is NFT?

It is a special type of token generated through cryptographic hashing methods and leverages blockchain technology for linking with a unique digital asset that could not be replicated. A non-fungible token is different from notable cryptocurrencies such as Monero, Ether, and Bitcoin in terms of fungibility. NFTs feature unique characteristics and cannot be exchanged or replaced with identical tokens.

Today, most NFTs are aesthetic. But there's a lot of innovation being done enabling dynamic utility of these assets that you can program.

Eric Anziani, COO of Crypto.com

NFT characteristics

1. Every single NFT is unique
2. Not replaceable with similar tokens
3. Uses ERC-721 and ERC-1155 token standards
4. The scarcity of NFT increases their value

Use cases of non-fungible token

- Digital art [Icona: digital-art.svg]
- Fashion [Icona: fashion.svg]
- Licenses and certifications [Icona: licenses.svg]
- Collectibles [Icona: collectibles.svg]
- Sports [Icona: sports.svg]
- Name service and domains [Icona: domains.svg]
- Virtual worlds [Icona: virtual-worlds.svg]
- Gaming [Icona: gaming.svg]

Standards in blockchains

Specific token standards have been created to support various blockchain use-cases. Ethereum was the first blockchain to support NFTs with its ERC-721 standard and is

currently the most widely used. Many other blockchains have added or plan to add support for NFTs with their growing popularity.

- **Ethereum** ERC-721 was the first standard for representing non-fungible digital assets on the Ethereum blockchain.
- **Others**
 1. **Bitcoin Cash**: Bitcoin Cash supports NFTs and powers the Juungle NFT marketplace.
 2. **Cardano**: Cardano introduced native tokens that enable the creation of NFTs without smart contracts with its March 2021 update. Cardano NFT marketplaces include CNFT and Theos.
 3. **Flow**: The Flow blockchain, which uses a proof of stake consensus model, supports NFTs. CryptoKitties plans to switch from Ethereum to Flow in the future.
 4. **GoChain**: GoChain, a blockchain which bills itself as 'eco-friendly', powers the Zeromint NFT marketplace and the VeVe app.
 5. **Solana**: The Solana blockchain also supports non-fungible tokens.
 6. **Tezos**: Tezos is a blockchain network that operates on proof of stake and supports the sale of NFT art.

buy.html

Where to buy NFTs

Opensea

[Aquí heu de situar la imatge opensea.png]

OpenSea is the **world's first and largest marketplace** where you can buy and sell NFTs. It's compatible with the Ethereum, Polygon, and Klaytn blockchains. You can browse through more than 34 million NFTs on this platform sorted into dozens of categories, ranging from digital art to collectibles, game items, virtual worlds, and domain names.

To sign up on OpenSea, you need an Ethereum wallet like MetaMask, Coinbase Wallet, or Fortmatic, and pay a signup fee to initialize your account. OpenSea also **takes a 2.5% cut** after the NFT has been sold. For fixed price items, the buyer pays the cut and for auctioned items, the seller pays it.

Axie Infinity

[Aquí heu de situar la imatge axie.png]

Axie Infinity is a gaming metaverse where users can create and breed pets called Axies and compete with one another in a card game. Each Axie is a **unique digital asset** with unique traits and can be bought or sold.

Axie Infinity is built on the Ethereum blockchain and requires a MetaMask wallet. It uses a pay-to-play-to-earn model where you can earn and cash out tokens (AXS or SLP) fortnightly as a reward. For all transactions you make, it **takes a 4.25% cut** from the buyer and the seller.

Rarible

[Aquí heu de situar la imatge rarible.png]

Rarible is another leading NFT marketplace for tokenized digital art. To sign up on the platform, **you must have an Ethereum wallet**. Rarible hosts a variety of NFT categories, including photography, music, digital art, domains, metaverses, and games. Perhaps the best aspect about this platform is that **you can earn royalties on the NFTs you resell**, which means you can create a stream of passive income and receive a cut every time the NFT gets sold.

Like most other marketplaces, Rarible **takes a 5% cut on every sale and splits it equally between the buyer and the seller**. That means both parties have to pay 2.5% of the final price.

SuperRare

[Aquí heu de situar la imatge superrare.png]

SuperRare is a **peer-to-peer digital art market that connects artists and collectors**. However, it is still in its early stages and so far, has only brought a handful of artists on board. SuperRare is unique in the sense that it is exclusive to only those artists who meet its criteria.

SuperRare only accepts payments in ETH and requires users to have a MetaMask wallet. You'll have to pay whatever the gas fee of the Ethereum network is at the time of buying. **It takes a 15% cut from the seller on every sale**, but also provides a 10% royalty each time the asset changes hands in the market. It also charges the buyer 3% of the NFT's final price.

Mintable

[Aquí heu de situar la imatge mintable.png]

Mintable is a gasless NFT marketplace where you can mint your memes, audio, or video files into NFTs. The platform is built on the Ethereum and Zilliqa blockchains, and **you need a MetaMask wallet to sign up on the platform**. Every time you buy an NFT, Mintable rewards you with a MINT token.

Like other NFT marketplaces, Mintable takes a cut on every sold item. But this cut isn't a flat fee. Instead, Mintable charges **2.5%, 5%, and 10%** for normal, gasless, and printable NFTs, respectively.

Capçalera

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Peu de pàgina

Student's full name. Subject: HTML & CSS

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Menú de navegació

- Article: Why We Keep Talking About NFTs [enllaça amb index.html]
- About NFTs [enllaça amb about.html]
- Where to buy [enllaça amb buy.html]

Enllaços dins dels arxius:

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- Pau Waelder [enllaça, dins del mateix arxiu, amb el fragment al final de la pàgina amb la trajectòria de l'autor de l'article]
- Library of Congress [Peu de la foto en blanc i negre. Enllaça amb <https://www.loc.gov/item/2014715740/>]
- “turning her website into an art gallery” [<http://art.teleportacia.org/exhibition/miniatures/>]
- Possession [<http://www.entropy8zuper.org/possession/>]
- [1] [enllaça, dins del mateix arxiu, amb el punt 1 de les notes al peu de l'article]
- [2] [enllaça, dins del mateix arxiu, amb el punt 2 de les notes al peu de l'article]
- [3] [enllaça, dins del mateix arxiu, amb el punt 3 de les notes al peu de l'article]
- [4] [enllaça, dins del mateix arxiu, amb el punt 4 de les notes al peu de l'article]
- ERC-721 standard [<https://ethereum.org/es/developers/docs/standards/tokens/erc-721/>]
- CryptoPunks [<https://www.larvalabs.com/cryptopunks>]
- CryptoKitties [<https://www.cryptokitties.co/>]
- OpenSea [<https://opensea.io/>]
- SuperRare [<https://superrare.com/>]
- Nifty Gateway [<https://niftygateway.com/>]
- Foundation [<https://foundation.app/>]
- Objkt [<https://objkt.com/>]
- Pace Verso [<https://verso.pacegallery.com/drops>]

- The Waiting Room [https://bitforms.art/exhibition/mark-napier-and-martin-wattenberg-interpreting-the-data/mn_waitingroom_1_w/]
- Altarboy [<http://zanni.org/index.php/portfolio/altarboy-oriana/>]
- “pieces designed to be seen on a web browser”
[<https://www.newrafael.com/websites/>]
- Github [<https://github.com/i3games/hen-timeline/blob/main/timeline.md>]

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